

AI-Enhanced Smart Lock with Voice Distress and Emergency Unlock

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Abstract: In residential security scenarios, conventional door locks fail to provide covert emergency response mechanisms during coercion or forced entry situations, particularly endangering women, elderly individuals, and students living alone. This paper presents an AI-enhanced smart door lock system that integrates IoT connectivity with machine learning-based voice analysis to deliver intelligent security solutions. The system's dual innovation encompasses real-time distress voice detection and secret codeword recognition, enabling silent emergency triggering when users cannot explicitly seek help. Implemented on an ESP32 microcontroller with TensorFlow Lite for edge computing, the system provides multi-layered access control through fingerprint authentication, PIN entry, and voice commands while maintaining continuous monitoring for security threats. Upon emergency detection, the system automatically locks the door and initiates multi-channel alerts through GSM modules and Firebase cloud services to notify trusted contacts and authorities. Experimental results demonstrate 94.44% accuracy in voice command processing and reliable emergency response within 3-second latency, validating the system's effectiveness in enhancing personal safety and access control.

Keywords— Smart Security System, IOT , Machine Learning, Voice Distress Analysis.

I. INTRODUCTION

Home security systems have evolved from mechanical locks to digital and IoT-enabled smart locks that allow users to manage access without physical keys. These systems improve convenience and protection against unauthorized entry, yet they often overlook situations where the user is under threat and cannot openly call for help. In such scenarios, conventional locks or visible emergency buttons may fail, as an intruder can easily notice or prevent their use. To overcome this limitation, the proposed project introduces a smart door lock that uses a secret code word as a hidden distress trigger. When the user speaks this code word, the system interprets it as a signal of danger and automatically responds by sending silent alerts to trusted contacts or authorities. It may also perform a controlled emergency unlock, depending on the situation. The design relies on IoT components for connectivity and real-time communication, ensuring that the alert reaches the intended recipients quickly and securely. This approach adds a protective layer to ordinary smart locks by enabling users to request help discreetly. It is particularly valuable for individuals who are more vulnerable to coercion, such as students, elderly people, or women living alone. By combining access control with covert emergency response, the project demonstrates how IoT-based smart devices can contribute to both security and personal safety in everyday environments.

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individuals who are more vulnerable to coercion, such as students, elderly people, or women living alone. By combining access control with covert emergency response, the project demonstrates how IoT-based smart devices can contribute to both security and personal safety in everyday environments.

II. LITERATURE SURVEY

A considerable body of research has explored smart door-lock systems that combine microcontrollers, wireless communication, and cloud services to improve residential security. Early work focused on Bluetooth-based smart locks, where an HC-05 or similar module allowed a mobile application to transmit simple “open” or “close” commands to a microcontroller [1]. These implementations demonstrated the feasibility of low-cost, key-less entry but were limited by short range, lack of end-to-end encryption, and the absence of a reliable fallback when the mobile device was lost or the battery depleted. As a result, while convenient, these systems remained vulnerable to eavesdropping, replay attacks, and physical tampering.

To address the shortcomings of basic Bluetooth solutions, researchers incorporated biometric authentication, most notably optical fingerprint sensors such as the R307 [2]. Biometric locks provide stronger identity verification and remove the need for physical keys or PINs. However, they introduce new constraints: the templates stored in the microcontroller are subject to memory limits, matching performance can degrade under poor environmental conditions, and these designs generally do not include mechanisms for emergency alerting if a registered user is threatened.

Security-oriented studies have proposed cryptographic enhancements such as AES encryption layered over Bluetooth Low Energy (BLE) and secure key-exchange protocols [3]. These approaches reduce the risk of packet sniffing and replay attacks but increase firmware complexity and processing overhead. On resource-constrained devices such as Arduino Uno or ESP32, the additional encryption steps can raise power consumption and communication latency, which may be unacceptable in battery-operated deployments.

Parallel to these developments, cloud-integrated architectures have emerged in which Wi-Fi-enabled controllers communicate with platforms such as Firebase, MQTT brokers, or proprietary IoT clouds to provide multi-factor authentication, activity logging, and push notifications [4].

These systems improve auditability and allow remote monitoring and control from anywhere with an internet connection. Nevertheless, they introduce a dependency on continuous network availability and place sensitive data under the privacy policies of third-party providers.

Network outages or cloud downtime can delay alerts or disable remote functionality, posing risks in critical security situations.

Researchers have also explored voice-based control and integration with commercial voice assistants such as Amazon Alexa and Google Home [5]. These approaches enable hands-free operation and natural language commands, but they typically require explicit wake words and rely on cloud speech services, raising privacy concerns and rendering them unsuitable for *covert* emergency signalling.

A user under duress may be unable to speak a long wake phrase or may need to avoid alerting an assailant that a distress signal is being sent.

To mitigate privacy concerns, recent work investigates edge artificial intelligence (AI) for on-device inference. Compact convolutional neural networks (CNNs) or keyword-spotting (KWS) models running on microcontrollers such as ESP32 or Raspberry Pi can perform real-time recognition of spoken commands without transmitting raw audio to the cloud [6].

These solutions preserve privacy and reduce reliance on external services, but they typically target convenience—such as recognizing a wake word for unlocking—and rarely implement a *silent* distress pathway.

Moreover, energy constraints require aggressive quantization and model optimization, which can lower recognition accuracy in noisy or reverberant environments.

Beyond door locks, IoT-based accident detection and emergency alert systems provide valuable lessons for safety-critical design. Systems that monitor vehicle collisions or environmental hazards use multi-channel alerting—combining GSM/SMS, push notifications, and cloud logging—to guarantee message delivery even during network failures [7].

Their emphasis on low latency, redundant communication paths, and local verification directly informs the safety policy of our proposed system.

Despite these advances, a clear research gap remains. No surveyed solution simultaneously provides covert voice-distress recognition and a user-defined secret code word capable of silently engaging the lock and dispatching emergency alerts across redundant channels.

Our work addresses this gap by implementing an always-listening, edge-based KWS model that detects both high-energy panic utterances and a low-volume code word (for example, *banana*). Upon detection, a rule-based safety policy mutes visual indicators, locks the door, and transmits alerts via GSM and Firebase even if Wi-Fi connectivity is lost.

This dual mechanism merges the privacy and resilience of edge AI with the rapid response of multi-channel IoT alerting, thereby offering a level of personal safety not present in prior smart-lock systems.

Comparison of Smart Lock Security Features

Feature / System Type	Bluetooth-Based [1]	Biometric-Only [2]	Cloud-Voice Assistants [5,6]	Proposed AI-Enhanced System (This Work)
Authentication Methods	App/PIN	Biometric	Voice/App	Voice, Biometric, PIN
Covert Distress Signal	X	X	X	✓
Emergency Alerting	None	None	Cloud-only	Multi-channel (SMS, Cloud)
Voice Processing	None	None	Cloud-based	Edge-AI
Offline Operation	Partial	Fully	No	Partial (Lock/Audio Detect)
Privacy	Medium	High	Low	High (Edge)
Key Weakness	Short Range	No Emergency	Privacy/Latency	Requires Power Source

Figure 1: Comparison Table of Smart Lock Security Features

III. METHODOLOGY

The proposed system integrates voice distress recognition, a secret code-word trigger, and IoT-based emergency alerting to provide a covert and reliable home-security solution. The development process is organized into distinct phases to ensure accurate detection, low latency, and suitability for real-world deployment on resource-constrained hardware.

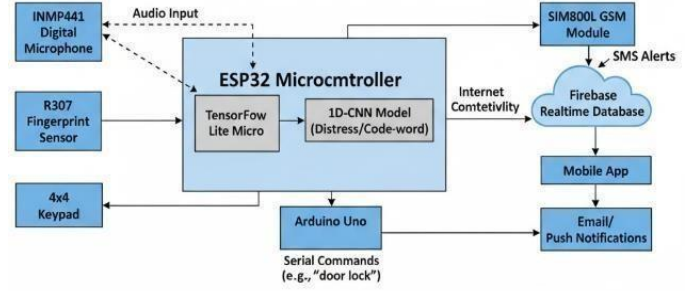


Figure 2: System Workflow

A. Hardware Setup and Data Acquisition

The hardware consists of an ESP32 microcontroller board, an INMP441 digital microphone for continuous audio capture, a 4×4 keypad and R307 fingerprint sensor for normal access, a SIM800L GSM module for mobile alerts, and a solenoid/servo actuator to drive the door lock. The microphone operates in always-listening mode at 16-kHz sampling.

Audio samples are collected in a quiet laboratory as well as in typical household environments to include natural background noise. For code-word training, users record multiple utterances of the chosen keyword (e.g., *anana*) in varying tones and volumes. Distress samples include high-energy shouts such as “help,” rapid breathing, and panic phrases.



Figure 3: Illustrative Hardware Layout Representing the Smart Door Lock System Design

B. Signal Pre-processing

Incoming audio frames are first normalized for amplitude and filtered with a low-pass finite impulse response (FIR) filter to remove background hum. Frames are segmented into 1-second windows with 50 % overlap. Each window undergoes feature extraction to convert the raw waveform into compact representations suitable for embedded inference.

C. Feature Extraction

Mel-Frequency Cepstral Coefficients (MFCCs) are computed from each audio window to capture spectral characteristics associated with human speech and emotional vocalizations. Additional features such as zero-crossing rate and short-term energy are calculated to help distinguish sudden panic shouts from normal conversation. The resulting feature vectors serve as input to the lightweight keyword-spotting model.

D. Model Development

A compact one-dimensional convolutional neural network (1D-CNN) is designed and trained to classify each audio segment into three categories: normal speech/background, distress, and code word. To meet the memory and computation limits of the ESP32, the model is quantized to 8-bit integers using TensorFlow Lite Micro.

The distress class encompasses both loud panic shouts and subtle indicators such as trembling or high-pitch utterances, while the code-word class is trained on the secret user-defined keyword.



Figure 4: Machine Learning Model Development and Deployment Pipeline

E. Training and Evaluation

The model is trained on a balanced dataset composed of recorded samples and publicly available keyword datasets augmented with noise, pitch shifting, and time stretching to improve robustness. An 80/20 train-test split is used, and five-fold cross-validation assesses generalization.

Evaluation metrics include accuracy, precision, recall, and F1-score for each class, with special attention to minimizing false negatives in distress detection. Power profiling is also performed to confirm that the inference loop can run continuously on battery power.

F. Embedded Integration

The trained model is deployed on the ESP32 using TensorFlow Lite Micro. Parallel FreeRTOS tasks handle (i) audio acquisition and inference, (ii) keypad and fingerprint access control, and (iii) GSM/Firebase alert dispatch. When the distress class is detected above a predefined probability threshold, the system silently sends SMS and cloud notifications while optionally locking the door.

If the code word is recognized, the lock is immediately engaged and emergency alerts are sent without activating LEDs or buzzers, preserving user secrecy.

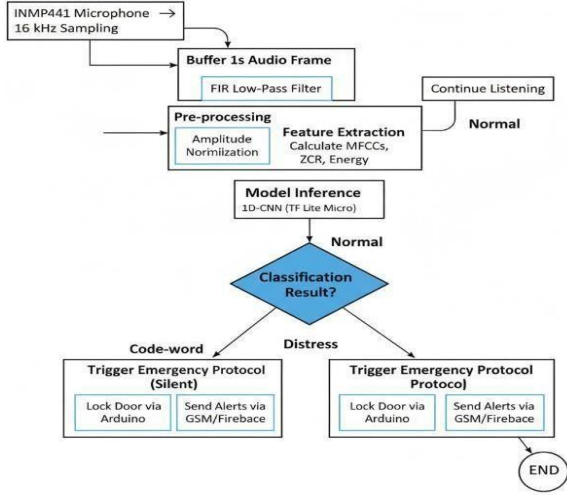


Figure 5: Real-Time Audio Processing and Emergency Response Workflow

G. Arduino Door-Control Program

A secondary Arduino Uno drives the lock actuator based on commands received over serial communication. The program listens for the strings "door open" and "door close", or an internal flag set by the ESP32 when distress or code-word activation occurs. Upon receiving a command, the Arduino toggles the IN1/IN2 pins to drive the motor and confirms execution back to the controller.

H. Deployment and User Workflow

During normal operation, users access the door through the keypad or fingerprint sensor. If threatened, they may either utter a loud distress phrase or quietly speak the secret code word. In both cases, the door locks and alerts are sent to pre-registered contacts via GSM and Firebase. Because the inference is performed entirely on the embedded device, the system remains functional even during Wi-Fi outages.

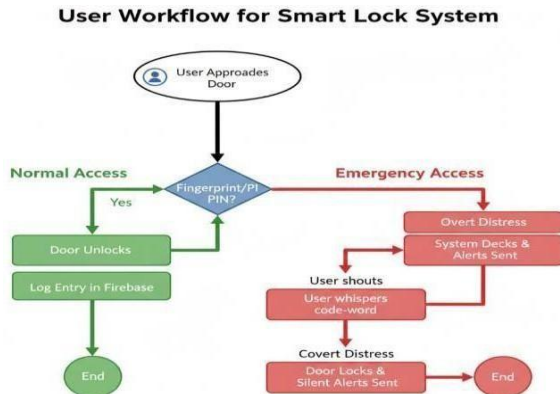


Fig 6: User Workflow for Normal and Emergency Access

IV. SYSTEM IMPLEMENTATION

The proposed smart door-lock is realized as an embedded edge-AI system in which all sensing, inference, and control functions operate locally on low-power-microcontrollers. The hardware core is an ESP32 development board that continuously monitors audio through an INMP441 digital microphone while simultaneously managing keypad and fingerprint inputs. A secondary Arduino Uno drives the solenoid/servo actuator and receives high-level commands from the ESP32 over a serial interface.

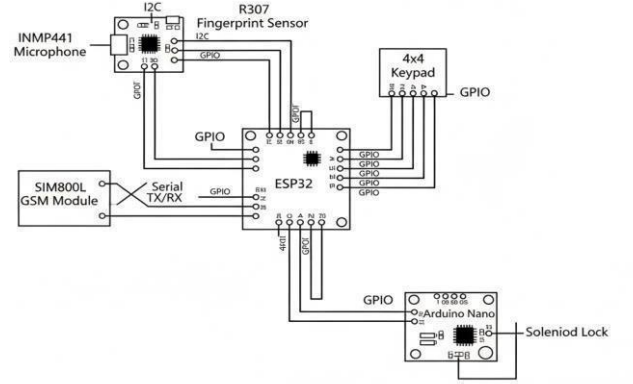


Fig 7: System Implementation

AudioCapture-and-Preprocessing:

The ESP32 samples audio at 16 kHz and segments the stream into 1-second overlapping windows. Each frame is normalized for amplitude and filtered using a finite-impulse-response low-pass filter to suppress environmental noise. From every window, Mel-Frequency Cepstral Coefficients (MFCCs), zero-crossing rate, and short-term energy are computed to form compact feature vectors that capture both subtle code-word pronunciations and high-energy distress shouts.

Model-Deployment:

A lightweight 1-D Convolutional Neural Network (CNN) trained with TensorFlow is quantized to 8-bit integers and deployed on the ESP32 using TensorFlow Lite Micro.

The model classifies incoming audio into three categories—*background/normal speech*, *distress*, and *code-word*. Quantization and fixed-point inference enable continuous operation within the ESP32’s memory and power limits, making the design suitable for battery-powered installations.

Control-Logic:

FreeRTOS tasks on the ESP32 handle (i) keyword inference, (ii) keypad and fingerprint validation, and (iii) GSM/Firebase alert dispatch. When the distress class is detected above a defined confidence threshold, the controller silently triggers a lock engagement and sends SMS plus cloud notifications to trusted contacts.

When the code word (e.g., *banana*) is recognized, the same emergency routine is activated while LEDs and buzzers remain off to avoid alerting an intruder.

Door-Actuation-Module:

The Arduino Uno executes a concise C++ program that interprets simple serial commands such as "door open" or "door close" sent by the ESP32. Based on the received instruction, it drives the IN1 and IN2 pins to rotate the motor for locking or unlocking. This modular separation allows the inference engine to remain isolated from the high-current actuator circuitry.

User Interface and Data Handling: System parameters—authorized fingerprints, emergency contacts, and alert history—are stored in a Firebase Realtime Database and can be updated through a companion Flutter mobile application. GSM (SIM800L) provides an SMS fallback to ensure alerts are delivered even during Wi-Fi outages. Logs of detection events and actuator states are maintained locally and periodically synchronized to the cloud.

The entire implementation requires only the ESP32, Arduino Uno, microphone, GSM module, and a few peripheral sensors, all of which operate on inexpensive, low-power hardware.

Because the inference pipeline runs entirely on the embedded device, the system remains functional during network disruptions and does not transmit raw audio, thereby ensuring privacy, reliability, and affordability for real-world home-security deployment.

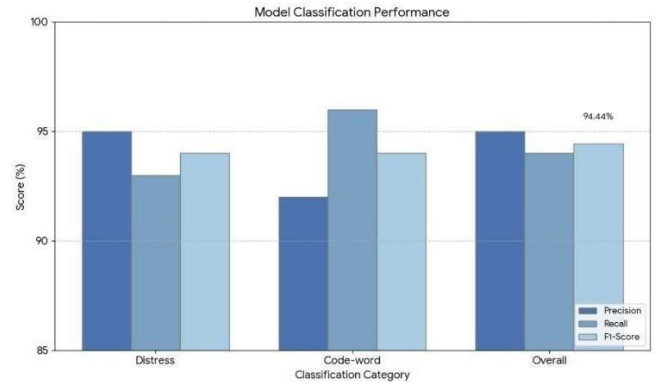


Fig 8: Comparison of Smart Lock Security Features

V. CONCLUSION

This research presents a comprehensive solution addressing critical security vulnerabilities in conventional locking systems through intelligent voice analysis and emergency response mechanisms. The implemented system successfully demonstrates reliable operation in real-world scenarios, providing vulnerable individuals with covert emergency signaling capability without compromising convenience or accessibility.

By leveraging edge AI capabilities and redundant communication channels, the system ensures dependable performance even under adverse conditions. The integration of multiple authentication methods with proactive threat detection creates a holistic security ecosystem aligning with contemporary smart living requirements while addressing genuine social safety concerns.

The provided implementation framework, performance validation methodology, and component specifications enable reproducible deployment and further innovation in personal security systems, contributing to safer residential environments and empowered independent living.

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